# Supplementary Information for: High Helium Retentivity in Goethite under Hydrothermal Conditions Measured by Step-Heat Experiments in Silica Glass Ampoules

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## 1. Analytical Methods for Hydrothermal Goethite

### *1.1. Uncertainty Quantification for Bounding Method*

*Helium-in-Goethite Diffusivity Inversions.* Apparent values of $D/a^2$ for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (**??**). Diffusivity for step $i$ is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^{i}_{SS}n_t\Big) \\ {}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\Big) \end{aligned} \tag{1}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield $\geq$7% (12 steps, Fig. **??**), yielding an $R^2$

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statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. **??**). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

*Helium Isotope Ratio Uncertainty.* The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotopes systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. **??**); 4He permeability itself could not be independently measured in this experiment (supplement, Helium-4 Blank Corrections). For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability ($\delta$) is taken to be 1.29, the $2\sigma$ upper bound of the value measured independently for a water-filled ampoule in Chapter **??** (Thyer et al., 2026): $1.10 \pm 0.19$ ($2\sigma$). 1.29 is in great excess of realistic values of the relative 3He/4He permeability, which is expected to be between 1.06-1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for

the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (**??**) (Chapter **??**): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$
2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$
3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$
4. $\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

Where $m_x^{i-1}$ and $(\lambda t)^{i-1}$ refer to the measured helium content and $\lambda t$ of the previous step, respectively. $\lambda$ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

*1.2. Finite Difference Diffusion Model of Cooling Tube*

One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in (Thyer et al., 2026). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°. Permeability values for 3He were assumed as the product

of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

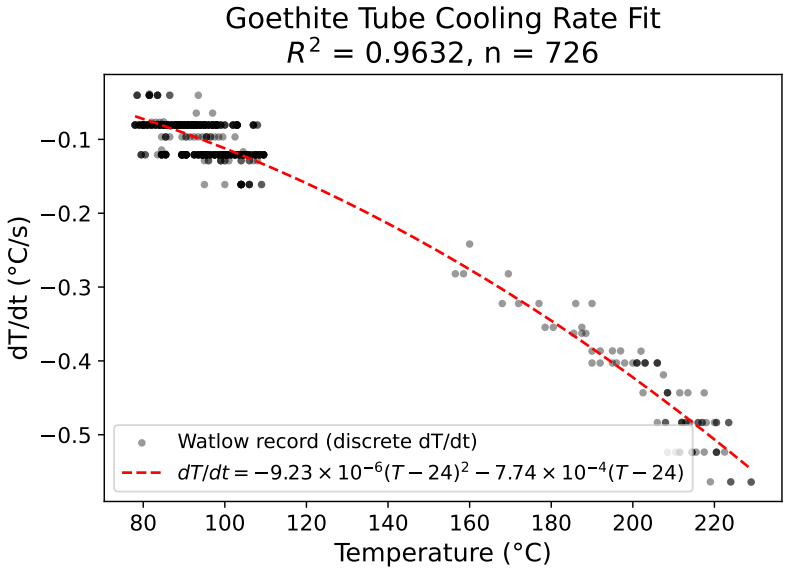



Figure S1: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

## 2. Verification of the Bounding Method for HT-Win-06-01b Using the Permeation Step Method

This section documents the permeation step method applied to this dataset: an independent, step-by-step treatment of the HT-Win-06-01b $^{3}$He dataset that explicitly tracks the helium content of the ampoule between steps, rather than bounding it as in the main text's approach (§**??**, §**??**). It is retained here solely as an internal verification of the bounding method used throughout this paper, not as the primary basis for the paper's results: the two methods' distilled $^{3}$He datasets and derived quantities agree, apart from the two "attempted complete degas" steps discussed in §**??**, corroborating the bounding method used in the main text. Because $^{4}$He permeation-step signals were blank level, no analogous direct check exists for $^{4}$He; $^{4}$He is therefore reduced exclusively by the bounding method throughout this paper and does not otherwise appear in this section. This section covers only the permeation step dataset and the step-by-step corrections applied to it; the isothermal-permeation-fit methodology and cooling-tube numerical model that establish the ampoule's elevated permeability – the evidentiary basis for the conservative literature-permeability assumption used in the main text, and a prerequisite for the corrections applied below – are described in §**??** and §1.2.

### *2.1. Summary*

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite of the Flinders Ranges (Monteiro, 2017), were acquired at Caltech on a Thermo Fisher Helix-MC-Plus$^{10K}$ mass spectrometer in peak hopping mode from January 31$^{st}$, 2024 to March 13$^{th}$, 2024. Raw data from these measurements are found on the first tab "Raw Data and Corrections" of the "IrrWin0601bTube012624FullDataBJT.xlsx" file. Because the sample was encapsulated in an imperfectly helium-transmissive silica glass ampoule, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C ) steps where the helium in goethite at high pH2O was effectively immobile for experimental timescales. For 3He, the resulting low-temperature measurements were above blank for the majority of the experiment and were entirely controlled by the process of permeation through silica glass, and therefore were used to estimate the amount of residual helium within the ampoule between high-temperature experimental steps. These corrections are carried out in the "Raw Data and Corrections" tab using permeability values derived internally within the

experiment from the "Naive Apparent Lambdas" tab. Diffusivities calculated using the equations of Fechtig and Kalbitzer (1966) for 3He and 4He can be found on the "Diffusivities" tab. During the step heat experiment, 116 measurements of high 3He MM gas standard were analyzed to track instrumental drift of 3He and 4He sensitivities and to quantify internal uncertainties using reproducibility. For the final step, the glass ampoule was broken and the remainder of the helium was degassed in a resistance furnace at 1100°C. Data from this final step can be found on the "Totals" tab of "IrrWin0601bTube012624FullDataBJT1.xlsx".

### *2.2. Raw Data and Corrections Columns*

#### *2.2.1. Raw Data*

"Nominal (Heated) Time" refers to the time programmed for the step heat. These times are used for both calculating apparent diffusivities with the equations of Fechtig and Kalbitzer (1966) and for permeation lag corrections.

The "Watlow Time" column gives the last timestamp recorded in the Watlow temperature log at the target temperature. It is always slightly higher than the nominal time because the recorded time includes the initial time required to heat the sample up to the programmed temperature: typically around 30 seconds to two minutes. This column was generated by a python script that read these values from the Watlow text file temperature records. It was used in early stages of data analysis to identify when Watlow errors within Labview had held the sample for longer than the programmed step duration. When these discrepancies were identified, the "Nominal Time" was updated to reflect the time the sample was held at the programmed temperature. These times are not directly used for any of the data reduction.

The "Total Time" column is the difference between the analysis timestamp and the previous analysis timestamp. These timestamps are recorded during the inlet of evolved, purified Helium into the Helix-MC-Plus$^{10K}$, which immediately precedes the pump out of the lamp cell chamber to the turbo pump. Therefore, the "Total Time" represents the total amount of time that helium was allowed to permeate through the silica glass ampoule into the UHV lamp cell, including both the time the sample spent "at temperature" and the time that the sample was cooling to or at room temperature (24-26°C). Since the pump out times were only 1 minute and always occurred when the sample was coolest (and permeation was slowest), permeation from the ampoule during pump out is neglected in the data reduction. Less than one thousandth of the helium remaining in the ampoule would permeate out

within one minute at room temperature (Eq. (**??**), with log permeability (ln(K)) at 26°C of 23.88 ln(atoms/cm*atm*s)). Due to the formatting of the time stamps, these values have a 1-minute resolution.

The "Cooling Time" is the difference between the "Total Time" and "Nominal Time". It is greater than or equal to the time to purify evolved gas in preparation for analysis: ~26 minutes. This value is used for permeation corrections when low-temperature permeation steps with cooling times greater than 26 minutes are used.

*2.2.2. Abundances, 4He Blank Corrections, Measured 4He/3He Sensitivity and Uncertainty Calculations.* 3He and 4He sensitivities were calculated for each analysis of the MM standard gas, assuming that each standard aliquot contained 157.50±0.41($2\sigma$) nccSTP 4He and 3.567±0.040($2\sigma$) pccSTP 3He. These values were obtained from sensitivities from a known amount of a standard gas with similar helium abundances and with the signals from 10 aliquots of the new MM standard tank in June 2023.

1 sigma uncertainties related to internal reproducibility were estimated for each sensitivity calculation as the rolling standard deviation of sensitivities calculated from 5 MM standards. In other words, the internal reproducibility of the sensitivity was estimated as the standard deviation of each calculated sensitivity and the 2 sensitivities immediately before and after. Uncertainties related to external reproducibility from the amount of 3He and 4He in each standard aliquot were added to the total uncertainty as well.

Sensitivities and their uncertainties were interpolated to times between standard analyses using a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) from the SciPy package in python. These interpolated sensitivities and uncertainties were used to convert the 3He and 4He signals (cps and fA respectively) to absolute atomic abundances and uncertainties in units of pccSTP and nccSTP respectively. The sensitivities and interpolations are graphically displayed in the "STDs" tab in a plot generated using the Matplotlib package in python.

*Helium-4 Blank Corrections.* Throughout the experiment, low temperature permeation steps yielded unpredictable 4He signals that did not exponentially decay like the 3He signals did. This effect is most obvious when considering analyses where cooling times were constant (~1560 seconds). In these cases, 3He signals always decreased exponentially whereas the 4He signals often increased (or decreased) erratically. A blank was measured at the

beginning of the experiment and showed negligible blank for 3He (regressing to below 0 cps at the inlet time) and a very low value for 4He ($4.84\text{x}10^{-2}$ fA). Low temperature permeation steps sometimes yielded much higher signals despite failing to conform to the expectation of exponential decay that is observed in the same steps' 3He measurements. For instance, the 4He signal in a cool permeation step at 105°C from reference file He30509.txt was $7.93\text{x}10^{0}$ fA, even when the high temperature experimental step preceding it (He30502.txt) yielded a similar value: $8.28\text{x}10^{0}$ fA. For the same pair of analyses, the 3He signal matched expectations and varied by two orders of magnitude. 4He from permeation steps never fit the equation for cumulative permeation, $m_{cum} = n_0 \left(1 - e^{-\lambda t}\right)$ (Eq. (**??**)), and thus could not be used to infer the permeability for 4He or for the residual amount of 4He remaining in the ampoule ($n_0$). For comparison, the same equation always fit the low temperature permeation steps well for 3He when the cooling time of each analysis was constant.

The erratic behavior in the 4He data is best explained by a shifting blank during the experiment, which was not measured or quantified directly. It was largest in the middle of the experiment and returned to low values by the end of the experiment that were similar to the blank measured before the experiment. Because the blank for 4He was not well controlled and appeared to have affected many of the values observed even in the high temperature experimental steps, a blank correction was deemed necessary. The 4He data of the permeation steps are interpreted here to be dominated by blanks, so high temperature experimental steps were coarsely blank corrected by subtracting the average 4He signal of the subsequent permeation steps. The entirety of this correction was conservatively added to the $2\sigma$ uncertainty of the experimental data for 4He. One blank corrected measurement was within error of zero after permeation corrections and was not plotted. The permeation lag was estimated for the 4He data using the 3He data and assuming that the 4He permeability was $0.91357 \pm 0.14603$ ($2\sigma$) that of the 3He permeability. These relative permeability values for 3He and 4He were measured at Caltech for a similar silica glass ampoule half-filled with water and equilibrated with $\sim 0.1$ torr of tank helium. This process is described in greater detail in the Permeation Lag Corrections section.

Some 3He permeation steps that followed especially low-yield experimental steps were within error of baseline 3He cps values (similarly to the measured blank at the beginning of the experiment). These low measurements always preceded by measurements where 3He was quantified above blank,

so permeation lag corrections were calculated using those values instead. In these cases, uncertainty from the permeation lag corrections dominate the uncertainty of the diffusivities, but do not affect the interpretation of the dataset.

*Bracketing Method Ratios.* 4He/3He ratios are presented with reference to the irradiated Winsor Goethite bulk ratio (~1000 or ~1000 Ra) and as standalone values. Uncertainties are propagated in quadrature from the measured 3He and 4He abundances and from the bulk ratio, when applicable. The blank and permeation corrected ratios have the same general shape as the uncorrected ratios. The ratios plateau around a cumulative 3He fraction of 0.06 at a bulk normalized 4He/3He ratio of 0.75. The early plateau in 4He/3He ratios with 3He fraction occurs similarly in the vacuum step heat experiments of the same sample, in agreement with previous studies (Monteiro, 2017; Farley et al., 2024), suggesting a relatively low loss of radiogenic helium for the Winsor Goethite. The vacuum and hydrothermal experiments differ in the bulk-normalized 4He/3He ratio at the plateau, with the hydrothermal experiment plateauing at a lower 4He/3He ratio of 0.75 times the bulk ratio, whereas the vacuum experiments plateau at the bulk ratio. This could be explained by a 4He-retentive inclusion in the hydrothermal experiment of this work similar to those discussed by Vasconcelos et al. (2013), or may reflect trapping of 3He in newly formed hematite during the early steps of the vacuum experiment.

### *2.2.3. Permeation Lag Corrections for 3He*

Helium evolved from the irradiated goethite must first permeate through the silica glass ampoule into the high vacuum of the lamp cell to be analyzed, a process which is non-instantaneous. Measurements can therefore underestimate the helium production of the sample during a step, as some of the produced helium will remain within the ampoule after a high temperature step, and measurements can also overestimate the amount of produced helium if there is some amount of helium remaining from previous step heats within the ampoule at the beginning of a new experimental step. To address this effect, we quantify the amount of helium remaining within the ampoule between high temperature experimental steps, and apply corrections using the rearranged form of Eq. (**??**), solved for $m$:

$$m = n_0 + \omega - n_t$$

Where $\omega$ is the true helium production of the sample in a high temperature experimental step, $n_0$ is the amount of helium within the ampoule at the beginning of the step, $n_t$ is the amount of helium within the ampoule at the end of the experiment, and $m$ is the amount measured. Values of $\omega$ and $m$ are most often within 10% of one another but differ by up to 47.2% for 3He within this dataset. $n_0$ and $n_t$ are tracked in the "3He $n_0$ (pccSTP)" column, where $n_t$ for any measurement is the $n_0$ value of the next row.

At low temperatures (<120°C), when no new helium is evolved from the sample, the measured value is simply the difference between the helium contents of the ampoule before the measurement and after (Eq. (**??**) with $\omega = 0$):

$$m = n_0 - n_t$$

In this case, all the helium measured is a result of helium permeation through the silica glass ampoule. Helium permeation is a first order process that follows Eq. (**??**), with $\lambda = KGRT$:

$$n_t = n_0 e^{-KGRTt}$$

Where $K$ is the temperature-dependent permeability coefficient, $G$ is a geometric factor which can be determined through measurements of the ampoule's inner and outer diameter, $R$ is the gas constant, $T$ is the temperature in Kelvin, and $t$ is the time in seconds. For a cylindrical ampoule, the value of G can be calculated with measurements of the inner and outer radii using Eq. (**??**). Inner and outer radii were measured using the amscope program at 11x zoom setting. These radii were both measured on opposing sides of the ampoule where there were air bubbles, allowing for the accurate sighting of the inner walls of ampoule. The inner diameter, *(2a)* was measured at 505 and 540 microns; the outer diameter *(2b)* was measured at 787 and 832 microns. The geometric factor $G$ was calculated for both pairs of measurements and averaged to give a value of $G$ of 6708.6 $cm^{-1}$.

For permeation steps, Eqs. (**??**) (with $\omega = 0$) and (**??**) apply because the temperature is too low to evolve any helium from the sample, and the $\omega = 0$ form of Eq. (**??**) can be substituted into Eq. (**??**) to relate a measurement to $n_0$. This equation is calculated in the column entitled "3He n0 calc (pccSTP) (m/(1-eˆ-λt))". Errors in the measured value $m$ and the permeability $K$ are propagated in quadrature. This is the permeation step relation of Eq. (**??**), expressed with Eq. (**??**)'s exponential form for ${}^{He}F$:

$$n_0 = \frac{m}{(1-e^{-KGRT})}$$

The "3He $n_0$ (pccSTP)" column tracks the accepted value of $n_0$ (and $n_t$, which is $n_0$ in the subsequent column) for ultimate use in permeation corrections using Eq. (**??**). Depending on the step, this value is determined one of three ways.

Most permeation steps in this study are inventory-setting measurements, which set $n_0$ directly from their own measurement using Eq. (**??**) – calculated in the "3He n0 calc (pccSTP) (m/(1-eˆ-λt))" column.

Permeation steps between inventory-setting measurements instead track the decreasing value of $n_0$ by the default first-order decay of Eq. (**??**): the calculated $n_0$ value for step $i$ is $n_0^i = n_0^{i-1} e^{-RG(KTt)_{i-1}}$. This default treatment is used whenever the previous step's cooling time matched the ~26-minute window used to calibrate the effective permeabilities themselves (§**??**), and is generally preferred over re-inverting from an inventory-setting measurement because it lets $n_0$ be set by earlier measurements with larger signals, avoiding error inflation from near-blank permeation steps. Errors associated with $K$ and $n_0^{i-1}$ are propagated through Eq. (**??**) in quadrature.

A small number of near-blank permeation steps, whose own signal was too close to baseline to invert reliably, instead had $n_0$ set by direct mass balance from an adjacent, reliably determined step: forward, by subtracting that step's own measured amount $m$ from the following step's $n_0$; or backward, by adding it to the preceding step's $n_0$.

Finally, some permeation steps and series – including the arbitrarily long "complete purge" steps at the end of the experiment discussed in §**??** – were long enough to be treated as emptying the ampoule entirely, so that $n_0$ for the following row was set to 0 rather than inverted from Eq. (**??**).

One further complication affects two adjacent high-temperature steps (195°C/1106 min and 185°C/740 min): the inventory-setting permeation step bracketing them cooled to room temperature and continued permeating for far longer (~13 hours) than the ~26-minute window used to calibrate every other step's effective permeability, and so would have needed a one-off, separately calibrated room-temperature permeability not used anywhere else in this reduction. Rather than build a special-case correction around two points, these two steps are instead excluded from the fitted Stage 1 regression (§2.3.1) and the diffusivity comparison of §**??** (Fig. S2), and marked (not derived) with "†" in Table **??** below: their differing cooling time, rather

than their signal strength, is the reason for the exclusion.

### *2.3. In-Situ Rate Constant Dataset*

The effective permeability coefficients used by the permeation step method below – and the isothermal-permeation-fit methodology, cooling-tube numerical model, and Arrhenius regression used to determine them – are described in full above (§**??**). The complete fitted dataset underlying that regression is reported here: Table S1 gives the raw per-step data for all 10 isothermal sequences, and Table S2 gives the lambda/$n_0$ fit to each sequence (Fig. **??**) that together define the Arrhenius regression of Fig. **??**.

Table S1: Raw step-heat data ('Fit Lambdas' sheet, cross-checked against 'Raw Data and Corrections') for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. **??**, Fig. **??**). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

| Series | Temp (°C) | Nominal Time (min) | Cooling Time (min) | $^3$He measured (pccSTP) | $\pm 2\sigma$ |
|---|---|---|---|---|---|
| 1 | 105.00 | 150 | 26 | 0.0190 | $1.1\times10^{-4}$ |
| 1 | 105.00 | 150 | 26 | 0.0111 | $7.4\times10^{-5}$ |
| 1 | 105.00 | 150 | 26 | 0.0080 | $5.3\times10^{-5}$ |
| 2 | 109.99 | 150 | 26 | 0.0229 | $1.4\times10^{-4}$ |
| 2 | 110.00 | 150 | 26 | 0.0115 | $7.3\times10^{-5}$ |
| 2 | 110.00 | 150 | 26 | 0.0075 | $5.2\times10^{-5}$ |
| 3 | 99.86 | 120 | 26 | 0.0106 | $6.1\times10^{-5}$ |
| 3 | 100.00 | 120 | 26 | 0.0074 | $4.0\times10^{-5}$ |
| 3 | 100.00 | 120 | 26 | 0.0049 | $2.5\times10^{-5}$ |
| 3 | 99.99 | 120 | 26 | 0.0043 | $2.1\times10^{-5}$ |
| 4 | 95.00 | 150 | 26 | 0.0109 | $2.8\times10^{-4}$ |
| 4 | 95.00 | 150 | 26 | 0.0081 | $2.0\times10^{-4}$ |
| 4 | 95.00 | 150 | 26 | 0.0062 | $1.4\times10^{-4}$ |
| 5 | 104.99 | 120 | 26 | 0.0215 | $8.9\times10^{-5}$ |
| 5 | 104.99 | 120 | 26 | 0.0144 | $6.0\times10^{-5}$ |
| 5 | 104.99 | 120 | 26 | 0.0104 | $4.3\times10^{-5}$ |
| 6 | 100.00 | 120 | 26 | 0.0126 | $5.6\times10^{-5}$ |
| 6 | 100.00 | 120 | 26 | 0.0094 | $3.5\times10^{-5}$ |
| 6 | 100.00 | 120 | 26 | 0.0074 | $2.0\times10^{-5}$ |
| 7 | 95.00 | 120 | 26 | 0.0072 | $3.0\times10^{-5}$ |
| 7 | 95.00 | 120 | 26 | 0.0063 | $2.6\times10^{-5}$ |

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*(continued)*

| Series | Temp (°C) | Nominal Time (min) | Cooling Time (min) | $^3$He measured (pccSTP) | $\pm 2\sigma$ |
|---|---|---|---|---|---|
| 7 | 95.00 | 120 | 26 | 0.0045 | $1.9\times10^{-5}$ |
| 8 | 90.00 | 150 | 26 | 0.0059 | $2.7\times10^{-5}$ |
| 8 | 89.90 | 150 | 26 | 0.0048 | $2.4\times10^{-5}$ |
| 8 | 90.00 | 150 | 26 | 0.0036 | $1.9\times10^{-5}$ |
| 9 | 90.00 | 180 | 26 | 0.0100 | $6.2\times10^{-5}$ |
| 9 | 89.92 | 180 | 26 | 0.0067 | $5.1\times10^{-5}$ |
| 9 | 90.00 | 180 | 26 | 0.0054 | $4.8\times10^{-5}$ |
| 10 | 99.99 | 150 | 26 | 0.0104 | $4.3\times10^{-5}$ |
| 10 | 99.98 | 150 | 26 | 0.0068 | $3.5\times10^{-5}$ |
| 10 | 100.00 | 150 | 26 | 0.0054 | $3.4\times10^{-5}$ |

Table S2: Fitted $\lambda$ (permeation rate constant) and $n_0$ (residual helium at the start of the series) for each of the 10 isothermal sequences in Table S1, from the weighted non-linear least-squares fit of Eq. (**??**) (Chapter **??**) described in the main text (§**??**). $n$ is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. **??** and used to build the Arrhenius regression in Fig. **??**.

| Series | Temp (°C) | Duration (min) | $n$ | $\lambda$ ($s^{-1}$) | $\pm 2\sigma$ | $n_0$ (pccSTP) | $\pm 2\sigma$ |
|---|---|---|---|---|---|---|---|
| 1 | 105.00 | 150 | 3 | $4.9\times10^{-5}$ | $1.4\times10^{-5}$ | 0.0521 | $7.8\times10^{-3}$ |
| 2 | 109.99 | 150 | 3 | $6.3\times10^{-5}$ | $1.8\times10^{-5}$ | 0.0508 | $7.3\times10^{-3}$ |
| 3 | 99.86 | 120 | 4 | $4.3\times10^{-5}$ | $1.3\times10^{-5}$ | 0.0377 | $5.2\times10^{-3}$ |
| 4 | 95.00 | 150 | 3 | $3.1\times10^{-5}$ | $2.6\times10^{-6}$ | 0.0444 | $2.2\times10^{-3}$ |
| 5 | 104.99 | 120 | 3 | $5.0\times10^{-5}$ | $6.2\times10^{-6}$ | 0.0696 | $4.7\times10^{-3}$ |
| 6 | 100.00 | 120 | 3 | $3.7\times10^{-5}$ | $4.1\times10^{-6}$ | 0.0534 | $3.4\times10^{-3}$ |
| 7 | 95.00 | 120 | 3 | $3.3\times10^{-5}$ | $1.5\times10^{-5}$ | 0.0348 | $1.1\times10^{-2}$ |
| 8 | 90.00 | 150 | 3 | $2.7\times10^{-5}$ | $4.8\times10^{-6}$ | 0.0279 | $3.5\times10^{-3}$ |
| 9 | 90.00 | 180 | 3 | $3.0\times10^{-5}$ | $9.0\times10^{-6}$ | 0.0357 | $7.0\times10^{-3}$ |
| 10 | 99.99 | 150 | 3 | $3.8\times10^{-5}$ | $1.2\times10^{-5}$ | 0.0351 | $7.0\times10^{-3}$ |

### *2.3.1. Permeation Step Method Step-Heat Dataset*

The complete corrected step-heat dataset underlying the permeation step method's apparent permeabilities (Fig. **??**) is reported in Table **??**, reporting the central value and $2\sigma$ uncertainty for each quantity, analogous to Table **??** in the main text for the bounding ("treatment as uncertainty") method. 4He is not treated by the permeation step method: 4He permeation-step signals were blank level (Helium-4 Blank Corrections), so no direct, step-by-step quantification of the ampoule's 4He inventory was possible, and 4He throughout this paper is instead reduced exclusively by the bounding method of the main text (§**??**). The permeation step dataset reported here is retained only for 3He, as the internal check on the bounding method discussed in §**??**. "–" denotes steps for which a quantity could not be determined (see §1.2 above).

Table S3: $^3$He step-heat dataset for HT-Win-06-01b under the naïve bracketing method: measured and cumulative $^3$He, the initial and residual ampoule content ($n_0$, $n_t$), the true step production ($\omega$), and the apparent diffusivity $\ln(D/a^2)$, each with its $2\sigma$ uncertainty. Six rows have derived values marked with "*": these steps have measured $^3$He $< 0.1$ pccSTP and are excluded as low-signal, as in Table **??**. Two of those six (195°C/1106 min and 185°C/740 min) are additionally marked "†": their $n_0$ depended on an atypically long room-temperature permeation rather than the standard cooling window used to calibrate the effective permeability (§**??**), so they are excluded (independently of the low-signal exclusion) from the fitted Stage 1 regression and Fig. S2.

| Temp (°C) | Time (min) | $^3$He meas. (pccSTP) | Cum. $^3$He (pccSTP) | $n_0$ (pccSTP) | $\pm2\sigma$ | $n_t$ (pccSTP) | $\pm2\sigma$ | $\omega$ (pccSTP) | $\pm2\sigma$ | $\ln(D/a^2)$ | $\pm2\sigma$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 170 | 300 | 0.9630 | 0.0000 | $0.00\times10^{0}$ | $0.00\times10^{0}$ | 0.1968 | $2.22\times10^{-2}$ | 1.1597 | $2.63\times10^{-2}$ | -21.910 | 0.045 |
| 180 | 463 | 0.5442 | 0.9630 | $7.26\times10^{-2}$ | $1.29\times10^{-2}$ | 0.0782 | $4.93\times10^{-3}$ | 0.5497 | $1.60\times10^{-2}$ | -22.184 | 0.089 |
| 190 | 550 | 0.5292 | 1.6241 | $1.95\times10^{-2}$ | $2.54\times10^{-3}$ | 0.0679 | $8.20\times10^{-3}$ | 0.5777 | $1.30\times10^{-2}$ | -21.976 | 0.085 |
| 200 | 500 | 0.5190 | 2.2112 | $3.28\times10^{-2}$ | $5.42\times10^{-3}$ | 0.0532 | $5.77\times10^{-3}$ | 0.5394 | $1.15\times10^{-2}$ | -21.702 | 0.088 |
| 210 | 475 | 0.5770 | 2.7646 | $1.41\times10^{-2}$ | $2.11\times10^{-3}$ | 0.0552 | $5.71\times10^{-3}$ | 0.6182 | $1.17\times10^{-2}$ | -21.311 | 0.082 |
| 220 | 450 | 0.6333 | 3.3799 | $1.10\times10^{-2}$ | $1.80\times10^{-3}$ | 0.0469 | $4.58\times10^{-3}$ | 0.6692 | $1.24\times10^{-2}$ | -20.991 | 0.080 |
| 230 | 425 | 0.7899 | 4.0551 | $5.87\times10^{-3}$ | $1.86\times10^{-3}$ | 0.0639 | $3.48\times10^{-3}$ | 0.8479 | $1.25\times10^{-2}$ | -20.514 | 0.066 |
| 225 | 908 | 0.7513 | 4.8885 | $1.95\times10^{-2}$ | $2.45\times10^{-3}$ | 0.0420 | $4.94\times10^{-3}$ | 0.7738 | $1.42\times10^{-2}$ | -21.200 | 0.121 |
| 215 | 450 | 0.1415 | 5.6842 | $1.31\times10^{-2}$ | $1.85\times10^{-3}$ | 0.0278 | $3.54\times10^{-3}$ | 0.1563 | $6.48\times10^{-3}$ | -22.015 | 0.316 |
| 205 | 520 | 0.0763 | 5.8529 | $7.18\times10^{-3*}$ | $1.23\times10^{-3*}$ | 0.0125* | $1.63\times10^{-3*}$ | 0.0816* | $8.10\times10^{-3*}$ | -22.788* | 0.829* |
| 195 | 1106 | 0.0670 | 5.9491 | $2.12\times10^{-3*\dagger}$ | $4.02\times10^{-4*\dagger}$ | 0.0307*† | $2.86\times10^{-3*\dagger}$ | 0.0956*† | $3.23\times10^{-3*\dagger}$ | -23.370*† | 1.094*† |
| 185 | 740 | 0.0227 | 6.0246 | $1.25\times10^{-2*\dagger}$ | $1.64\times10^{-3*\dagger}$ | 0.0090*† | $1.06\times10^{-3*\dagger}$ | 0.0192*† | $2.00\times10^{-3*\dagger}$ | -24.567*† | 2.822*† |
| 175 | 920 | 0.0180 | 6.0610 | $3.68\times10^{-3*}$ | $5.02\times10^{-4*}$ | 0.0045* | $4.42\times10^{-4*}$ | 0.0189* | $7.42\times10^{-4*}$ | -24.796* | 3.797* |
| 200 | 260 | 0.0224 | 6.0814 | $6.53\times10^{-4*}$ | $1.17\times10^{-4*}$ | 0.0095* | $1.23\times10^{-3*}$ | 0.0313* | $1.65\times10^{-3*}$ | -23.022* | 1.495* |
| 210 | 520 | 0.0812 | 6.1081 | $7.48\times10^{-3*}$ | $9.98\times10^{-4*}$ | 0.0145* | $1.32\times10^{-3*}$ | 0.0882* | $3.85\times10^{-3*}$ | -22.669* | 1.135* |
| 220 | 1167 | 0.5766 | 6.1914 | $2.82\times10^{-3}$ | $1.44\times10^{-3}$ | 0.0276 | $3.24\times10^{-3}$ | 0.6014 | $2.45\times10^{-2}$ | -21.504 | 0.184 |
| 230 | 370 | 0.3577 | 6.7796 | $8.68\times10^{-3}$ | $3.96\times10^{-3}$ | 0.0422 | $5.13\times10^{-3}$ | 0.3912 | $1.48\times10^{-2}$ | -20.712 | 0.146 |
| 240 | 320 | 0.6767 | 7.1563 | $1.72\times10^{-2}$ | $2.42\times10^{-3}$ | 0.0826 | $9.94\times10^{-3}$ | 0.7420 | $1.57\times10^{-2}$ | -19.807 | 0.099 |
| 245 | 250 | 0.5549 | 7.8582 | $2.13\times10^{-2}$ | $8.87\times10^{-3}$ | 0.0516 | $5.65\times10^{-3}$ | 0.5852 | $1.46\times10^{-2}$ | -19.710 | 0.125 |
| 250 | 210 | 0.6934 | 8.4743 | $1.08\times10^{-2}$ | $4.52\times10^{-3}$ | 0.0718 | $8.15\times10^{-3}$ | 0.7545 | $1.41\times10^{-2}$ | -19.200 | 0.102 |

*continued on next page*

*(continued)*

| Temp (°C) | Time (min) | $^3$He meas. (pccSTP) | Cum. $^3$He (pccSTP) | $n_0$ (pccSTP) | $\pm 2\sigma$ | $n_t$ (pccSTP) | $\pm 2\sigma$ | $\omega$ (pccSTP) | $\pm 2\sigma$ | $\ln(D/a^2)$ | $\pm 2\sigma$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 245 | 270 | 0.4659 | 9.2086 | $2.48\times10^{-2}$ | $3.48\times10^{-3}$ | 0.0498 | $5.84\times10^{-3}$ | 0.4909 | $1.01\times10^{-2}$ | -19.810 | 0.197 |
| 240 | 320 | 0.2857 | 9.7206 | $2.07\times10^{-2}$ | $2.81\times10^{-3}$ | 0.0339 | $4.08\times10^{-3}$ | 0.2989 | $6.58\times10^{-3}$ | -20.433 | 0.312 |
| 235 | 370 | 0.2199 | 10.0357 | $1.66\times10^{-2}$ | $2.20\times10^{-3}$ | 0.0273 | $3.29\times10^{-3}$ | 0.2307 | $5.23\times10^{-3}$ | -20.810 | 0.401 |
| 230 | 420 | 0.1400 | 10.2736 | $1.32\times10^{-2}$ | $1.76\times10^{-3}$ | 0.0255 | $2.53\times10^{-3}$ | 0.1523 | $4.00\times10^{-3}$ | -21.332 | 0.603 |
| 225 | 652 | 0.1494 | 10.4279 | $1.26\times10^{-2}$ | $1.62\times10^{-3}$ | 0.0131 | $1.47\times10^{-3}$ | 0.1499 | $3.42\times10^{-3}$ | -21.772 | 0.747 |
| 230 | 500 | 0.1550 | 10.5908 | $7.34\times10^{-4}$ | $1.67\times10^{-3}$ | 0.0128 | $1.32\times10^{-3}$ | 0.1671 | $3.37\times10^{-3}$ | -21.383 | 0.461 |
| 235 | 450 | 0.1951 | 10.7582 | $2.57\times10^{-3}$ | $4.18\times10^{-4}$ | 0.0334 | $1.85\times10^{-3}$ | 0.2259 | $3.29\times10^{-3}$ | -20.956 | 0.364 |
| 240 | 400 | 0.2425 | 10.9638 | $6.90\times10^{-3}$ | $1.45\times10^{-3}$ | 0.0251 | $3.16\times10^{-3}$ | 0.2607 | $1.23\times10^{-2}$ | -20.672 | 0.318 |
| 245 | 350 | 0.3150 | 11.2328 | $5.79\times10^{-3}$ | $1.05\times10^{-3}$ | 0.0395 | $4.64\times10^{-3}$ | 0.3487 | $9.39\times10^{-3}$ | -20.219 | 0.240 |
| 250 | 300 | 0.3716 | 11.5689 | $1.65\times10^{-2}$ | $2.24\times10^{-3}$ | 0.0467 | $5.24\times10^{-3}$ | 0.4018 | $9.62\times10^{-3}$ | -19.888 | 0.209 |
| 250 | 300 | 0.3640 | 11.9627 | $2.43\times10^{-2}$ | $4.37\times10^{-3}$ | 0.0339 | $3.82\times10^{-3}$ | 0.3735 | $7.99\times10^{-3}$ | -19.927 | 0.242 |
| 250 | 300 | 0.3452 | 12.3490 | $1.13\times10^{-2}$ | $1.60\times10^{-3}$ | 0.0395 | $4.48\times10^{-3}$ | 0.3733 | $7.91\times10^{-3}$ | -19.895 | 0.244 |
| 250 | 10250 | 8.4829 | 12.7167 | $1.91\times10^{-2}$ | $2.53\times10^{-3}$ | 0.1165 | $7.38\times10^{-3}$ | 8.5804 | $1.28\times10^{-1}$ | -19.958 | 0.189 |
| 240 | 785 | 0.2261 | 21.2187 | $2.90\times10^{-9}$ | $6.85\times10^{-9}$ | 0.0263 | $2.92\times10^{-3}$ | 0.2525 | $2.21\times10^{-2}$ | -20.660 | 0.580 |
| **Total** | | | 146.1483 | | | | | | | | |

Beyond the ampoule's residual helium content itself (§**??**), the two methods' derived apparent diffusivities agree just as closely. Fig. S2 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own "treatment as uncertainty" spread on one axis and the permeation-step method's propagated uncertainty on the other; the two steps marked "†" in Table **??** above are excluded for the reason given there. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. S3) yields an activation energy of 157.0±4.8 kJ/mol ($R^2$=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (§**??**).

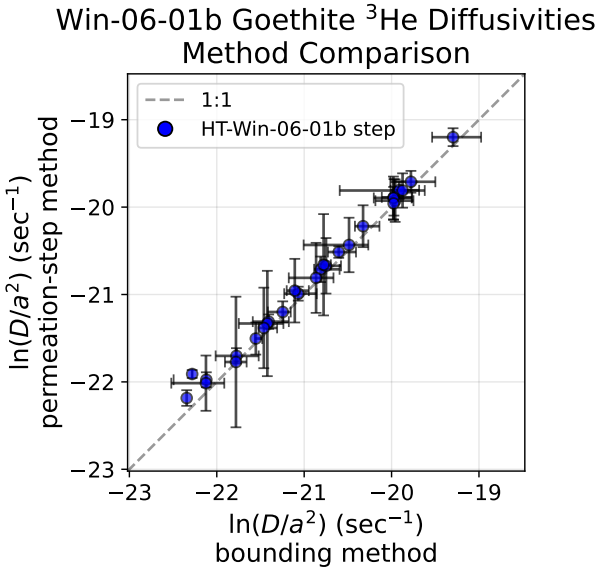



Figure S2: Apparent $^3$He diffusivities for HT-Win-06-01b from the permeation-step method ($y$-axis) against the bounding method's central value ($x$-axis, Eq. (**??**)), with the bounding method's "treatment as uncertainty" LB/UB spread and the permeation-step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1. Two steps (195°C/1106 min and 185°C/740 min) are excluded for the reason given in Table **??**'s caption.

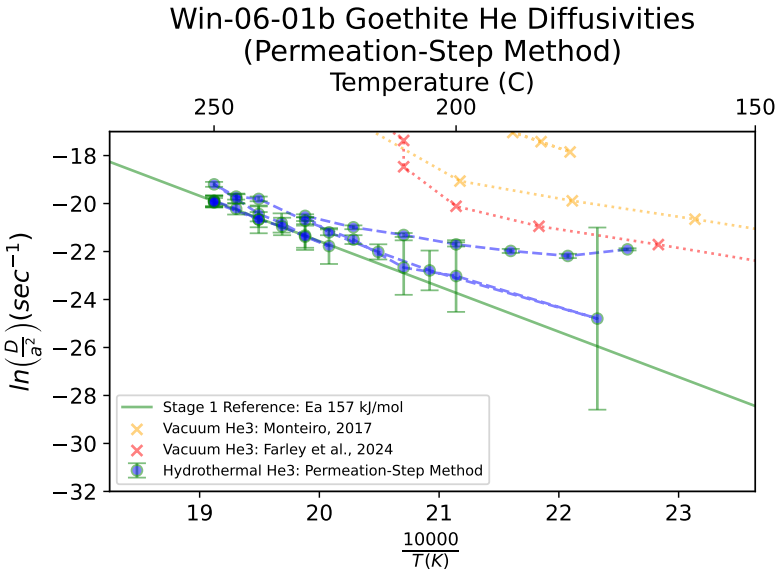



Figure S3: Analogue of Fig. **??** (main text) using the permeation-step method's own $^3$He diffusivities and its own Stage 1 linear regression, rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C).

The finite-difference numerical model used to evaluate the contribution of the 26-minute cooling period to these apparent permeabilities – and to confirm that the ampoule's markedly elevated permeability is not an artifact of that cooling history – is described in full above (§1.2).

*2.4. Diffusivities*

Apparent values of $D/a^2$ for 3He and 4He were calculated from times and corrected cumulative yields using the equations of Fechtig and Kalbitzer (1966). Uncertainties in the corrected values are propagated into the cumulative yield, and all errors are treated as uncorrelated for simplicity. Error bars plotted in the Arrhenius plots on this tab are likely exaggerated. Derivatives necessary to calculate uncertainty in the diffusivities using quadrature are tabulated to the right of the sheet. The Stage 1 reference-line linear regression (Fig. **??**), its statistics, and the treatment of total-yield uncertainty in the regression are described in the main text (§**??**); the same regression underlies both the main text's bounding-method results and the permeation step dataset reported here.

*2.5. Totals*

The main text (§**??**) already summarizes this final furnace-fusion step and its blank/re-extract corrections; the additional spreadsheet-level detail below (volume-split calibration, standard-averaging convention) is retained here for completeness. For the final step, the glass ampoule was broken and the remainder of the helium was completely degassed in a resistance furnace at 1100°C. The gas from this final step was split within the vacuum line on calibrated volumes so that purified helium from less than 1% of the evolved gas was inlet into a Thermo Fisher Helix SFT mass spectrometer. A gas standard with a known amount of 3He and 4He was measured twice preceding the final fusion measurement of the hydrothermal Win-06-01b, as was a split furnace hot blank. The gas standards were used to calculate sensitivities, which were taken to be the mean of the two measurements, with $1\sigma$ uncertainty approximated by their standard deviation. After the split measurement of the Win-06-01b ampoule, the furnace was reheated to 1100°C to ensure complete extraction of the helium in the sample. The final step was then taken to be the measured value minus the hot blank, plus the re-extract, all divided by their respective volume split fractions and multiplied by the calculated sensitivity. Uncertainties were propagated in quadrature.

## 3. Encapsulated Durango apatite step-heat data

The following table gives the complete step-heat dataset for the silica glass encapsulated Durango apatite experiment described in §**??** and plotted in Fig. **??**. Permeation lag corrections follow the permeation-step route of §**??**: Eq. (**??**) is applied to the permeation steps shown in **bold** to solve for the ampoule's helium content, and Eq. (**??**) propagates that content forward and backward in time to the neighbouring experimental steps.

Only the last of each sequence of room-temperature measurements was used to solve for $n_0$, because the temperature records show that the ampoule had not yet cooled to room temperature when the first permeation step of each sequence began. Each pair of permeation steps nonetheless yielded values of $n_0$ within $2\sigma$ of one another.

Measurement uncertainties are estimated from the standard deviations of sensitivities and blanks determined by standards, and are propagated through Eqs. (**??**) and (**??**) in quadrature, together with a general 10% uncertainty on the use of Eq. (**??**). Uncertainty on a step's cumulative fractional yield is the sum of the uncertainty on its own fractional yield and those of all preceding steps, added in quadrature. The total yield is estimated from the measured mass of encapsulated apatite (3.18 mg) and its helium concentration of 194.2 nccSTP/mg (Farley, 2000); no uncertainty on the total yield is propagated into the reported diffusivities.

Table S4: Step heat data for the silica glass encapsulated Durango apatite experiment. $^{He}F_{Tube}$ is the fractional permeation yield (Eq. (**??**)) of the bracketing permeation step; $^{He}F_{Apatite}$ is the permeation lag-corrected cumulative fractional yield of helium released from the apatite. $n_0$ values (initial $^4$He content of the ampoule, Eq. (**??**)) are shown in **bold** for permeation steps used directly to solve for $n_0$, and in *italics* for all other steps, where $n_0$ is instead propagated forward or backward in time from the nearest bolded permeation step using Eq. (**??**). Blank "Type" entries denote room-temperature measurements preceding the permeation step used to solve for $n_0$, taken before the diffusive steady state concentration profile was reestablished (see text). Uncertainties are $2\sigma$. Corrected $\ln(D/a^2)$ values shown as "–" correspond to steps where the corrected apatite yield ($\omega$) was within $2\sigma$ of 0 and were excluded from the Arrhenius plot (Fig. **??**).

| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | $^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | $\omega$ (nccSTP $^4$He) | $^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| Permeation | 15 | 24.3 | 0.00704 | ± 4.76‰ | **1.48** | ± | | | |
| | | | 0.00046 | | **0.18** | | | | |
| Experimental | 240 | 100.0 | 0.1700 ± 0.0011 | | *1.47* | ± 0.05 | ± 0.00803% | – | -28.39 |
| | | | | | *0.18* | 0.24 | | | |
| | 15 | 28.0 | 0.00721 | ± | *1.35* | ± | | | |
| | | | 0.00039 | | *0.16* | | | | |
| | 15 | 24.0 | 0.00639 | ± | *1.34* | ± | | | |
| | | | 0.00039 | | *0.16* | | | | |
| Permeation | 15 | 24.0 | 0.00630 | ± 4.71‰ | **1.34** | ± | | | |
| | | | 0.00039 | | **0.16** | | | | |
| Experimental | 240 | 150.0 | 0.3090 ± 0.0023 | | *1.33* | ± 0.01 | ± 0.00983% | – | -26.98 |
| | | | | | *0.16* | 0.21 | | | |
| | 15 | 28.1 | 0.00542 | ± | *1.03* | ± | | | |
| | | | 0.00039 | | *0.13* | | | | |
| | 15 | 24.3 | 0.00484 | ± | *1.03* | ± | | | |
| | | | 0.00039 | | *0.13* | | | | |

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*(continued)*

| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | | ${}^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | | $\omega$ (nccSTP $^4$He) | ${}^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Permeation | 15 | 24.2 | 0.00485 | ± | 4.73‰ | **1.02** | ± | | | | |
| | | | 0.00039 | | | **0.13** | | | | | |
| Experimental | 240 | 200.1 | 0.9416 ± 0.0085 | | | *1.02* | ± | 1.64 | ± 0.276% | -23.80 | ± -25.02 |
| | | | | | | *0.13* | | 0.23 | | 0.28 | |
| | 15 | 29.9 | 0.00947 | ± | | *1.72* | ± | | | | |
| | | | 0.00041 | | | *0.19* | | | | | |
| | 15 | 23.8 | 0.00786 | ± | | *1.71* | ± | | | | |
| | | | 0.00040 | | | *0.19* | | | | | |
| Permeation | 15 | 24.0 | 0.00804 | ± | 4.72‰ | **1.70** | ± | | | | |
| | | | 0.00040 | | | **0.19** | | | | | |
| Experimental | 60 | 250.2 | 1.097 ± 0.011 | | | *1.69* | ± | 3.70 | ± 0.874% | -20.21 | ± -24.01 |
| | | | | | | *0.19* | | 0.48 | | 0.20 | |
| | 15 | 33.0 | 0.02653 | ± | | *4.30* | ± | | | | |
| | | | 0.00050 | | | *0.44* | | | | | |
| | 15 | 23.5 | 0.02016 | ± | | *4.27* | ± | | | | |
| | | | 0.00046 | | | *0.44* | | | | | |
| Permeation | 15 | 24.0 | 0.02003 | ± | 4.72‰ | **4.25** | ± | | | | |
| | | | 0.00047 | | | **0.44** | | | | | |
| Experimental | 60 | 270.3 | 2.508 ± 0.027 | | | *4.23* | ± | 5.26 | ± 1.73% | -19.04 | ± -22.51 |
| | | | | | | *0.44* | | 0.83 | | 0.22 | |
| | 15 | 33.9 | 0.04414 | ± | | *6.98* | ± | | | | |
| | | | 0.00067 | | | *0.70* | | | | | |

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*(continued)*

| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | $^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | $\omega$ (nccSTP $^4$He) | $^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| Permeation | 15 | 23.6 | 0.03246 ± 0.00057 | ± 4.68‰ | **6.93 ± 0.70** | | | | |
| Experimental | 10 | 290.3 | 4.979 ± 0.058 | | *6.90 ± 0.70* | 9.0 ± 1.3 | 3.18% | -17.87 ± 0.20 | -21.10 |
| | 15 | 33.6 | 0.06816 ± 0.00095 | | *10.9 ± 1.1* | | | | |
| Permeation | 15 | 23.6 | 0.05050 ± 0.00078 | ± 4.66‰ | **10.8 ± 1.1** | | | | |
| Experimental | 10 | 300.3 | 7.243 ± 0.091 | | *10.8 ± 1.1* | 9.5 ± 1.7 | 4.72% | -17.34 ± 0.25 | -20.07 |
| | 15 | 34.2 | 0.0843 ± 0.0012 | | *13.1 ± 1.3* | | | | |
| | 15 | 23.5 | 0.06160 ± 0.00094 | | *13.0 ± 1.3* | | | | |
| Permeation | 15 | 24.1 | 0.06097 ± 0.00095 | ± 4.72‰ | **12.9 ± 1.3** | | | | |
| Experimental | 10 | 285.3 | 5.919 ± 0.080 | | *12.9 ± 1.3* | 3.6 ± 1.7 | 5.30% | -18.04 ± 0.71 | -20.65 |
| | 15 | 36.8 | 0.0729 ± 0.0011 | | *10.5 ± 1.1* | | | | |
| Permeation | 15 | 23.6 | 0.04846 ± 0.00084 | ± 4.64‰ | **10.4 ± 1.1** | | | | |
| Experimental | 10 | 275.2 | 5.382 ± 0.077 | | *10.4 ± 1.1* | 2.5 ± 1.3 | 5.71% | -18.6 ± 1.1 | -19.80 |
| | 15 | 37.0 | 0.05243 ± 0.00092 | | *7.53 ± 0.76* | | | | |

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*(continued)*

| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | | ${}^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | | $\omega$ (nccSTP $^4$He) | | ${}^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Permeation | 15 | 23.6 | 0.03487 | ± | 4.66‰ | **7.48** | ± | | | | | |
| | | | 0.00070 | | | **0.76** | | | | | | |
| Experimental | 100 | 255.2 | 3.382 ± 0.052 | | | *7.44* | ± | 0.60 | ± | 5.81% | – | -20.22 |
| | | | | | | *0.76* | | 0.90 | | | | |
| | 15 | 37.1 | 0.03206 | ± | | *4.66* | ± | | | | | |
| | | | 0.00069 | | | *0.48* | | | | | | |
| Permeation | 15 | 23.6 | 0.02147 | ± | 4.64‰ | **4.63** | ± | | | | | |
| | | | 0.00056 | | | **0.48** | | | | | | |
| Experimental | 60 | 300.3 | 3.684 ± 0.059 | | | *4.61* | ± | 6.68 | ± | 6.89% | -17.17 | ± -20.76 |
| | | | | | | *0.48* | | 0.91 | | | 0.20 | |
| | 15 | 36.3 | 0.0518 ± 0.0010 | | | *7.60* | ± | | | | | |
| | | | | | | *0.77* | | | | | | |
| Permeation | 15 | 23.6 | 0.03513 | ± | 4.65‰ | **7.55** | ± | | | | | |
| | | | 0.00077 | | | **0.77** | | | | | | |
| Experimental | 120 | 300.2 | 9.78 ± 0.17 | | | *7.51* | ± | 11.1 | ± | 1.2 8.69% | -17.13 | ± -19.21 |
| | | | | | | *0.77* | | | | | 0.22 | |
| | 15 | 35.7 | 0.0595 ± 0.0012 | | | *8.85* | ± | | | | | |
| | | | | | | *0.90* | | | | | | |
| Permeation | 15 | 23.6 | 0.04087 | ± | 4.65‰ | **8.79** | ± | | | | | |
| | | | 0.00089 | | | **0.90** | | | | | | |
| Experimental | 180 | 300.1 | 14.58 ± 0.27 | | | *8.75* | ± | 13.8 | ± | 1.2 10.9% | -17.08 | ± -18.75 |
| | | | | | | *0.90* | | | | | 0.18 | |

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| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | $^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | $\omega$ (nccSTP $^4$He) | $^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 35.9 | 0.0547 ± 0.0012 | | *7.96 ± 0.81* | | | | |
| Permeation | 15 | 23.6 | 0.03782 ± 0.00089 | ± 4.78‰ | **7.91 ± 0.81** | | | | |
| Experimental | 10 | 320.3 | 7.49 ± 0.14 | | *7.87 ± 0.81* | 11.4 ± 1.5 | 12.8% | -15.96 ± 0.18 | -17.94 |
| | 15 | 36.9 | 0.0813 ± 0.0017 | | *11.7 ± 1.2* | | | | |
| Permeation | 15 | 23.5 | 0.0541 ± 0.0012 | 4.64‰ | **11.7 ± 1.2** | | | | |
| Experimental | 10 | 330.4 | 11.18 ± 0.22 | | *11.6 ± 1.2* | 14.5 ± 1.9 | 15.1% | -15.54 ± 0.21 | -18.08 |
| | 15 | 37.4 | 0.1076 ± 0.0023 | | *14.9 ± 1.5* | | | | |
| Permeation | 15 | 23.7 | 0.0691 ± 0.0016 | 4.67‰ | **14.8 ± 1.5** | | | | |
| Experimental | 10 | 350.4 | 19.78 ± 0.41 | | *14.7 ± 1.5* | 28.3 ± 2.8 | 19.7% | -14.61 ± 0.15 | -18.11 |
| | 15 | 37.0 | 0.1615 ± 0.0035 | | *23.3 ± 2.4* | | | | |
| Permeation | 15 | 23.5 | 0.1071 ± 0.0024 | 4.64‰ | **23.1 ± 2.4** | | | | |
| Experimental | 10 | 370.4 | 32.97 ± 0.71 | | *23.0 ± 2.4* | 41.3 ± 4.0 | 26.4% | -13.90 ± 0.15 | -16.15 |
| | 15 | 37.3 | 0.2207 ± 0.0049 | | *31.4 ± 3.2* | | | | |

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| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | $^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | $\omega$ (nccSTP $^4$He) | $^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| Permeation | 15 | 23.5 | 0.1447 ± 0.0033 | 4.65‰ | **31.1 ± 3.2** | | | | |
| Experimental | 60 | 385.4 | 42.32 ± 0.94 | | *31.0 ± 3.2* | 44.8 ± 4.8 | 33.6% | -13.48 ± 0.17 | -15.41 |
| | 15 | 37.8 | 0.2393 ± 0.0055 | | *33.5 ± 3.4* | | | | |
| Permeation | 15 | 23.5 | 0.1548 ± 0.0037 | 4.66‰ | **33.2 ± 3.4** | | | | |
| Experimental | 60 | 360.4 | 27.53 ± 0.64 | | *33.1 ± 3.4* | 15.3 ± 4.1 | 36.1% | -14.35 ± 0.51 | -15.33 |
| | 15 | 39.6 | 0.1620 ± 0.0039 | | *20.9 ± 2.1* | | | | |
| Permeation | 15 | 23.6 | 0.0963 ± 0.0024 | 4.65‰ | **20.7 ± 2.1** | | | | |
| Experimental | 60 | 340.4 | 14.78 ± 0.35 | | *20.6 ± 2.1* | 7.4 ± 2.6 | 37.3% | -15.00 ± 0.90 | -15.63 |
| | 15 | 40.0 | 0.1029 ± 0.0026 | | *13.2 ± 1.4* | | | | |
| Permeation | 15 | 23.5 | 0.0610 ± 0.0017 | 4.64‰ | **13.1 ± 1.4** | | | | |
| Experimental | 100 | 325.2 | 12.37 ± 0.31 | | *13.1 ± 1.4* | 6.8 ± 1.6 | 38.4% | -15.6 ± 1.1 | -13.69 |
| | 15 | 40.1 | 0.0589 ± 0.0017 | | *7.50 ± 0.78* | | | | |
| Permeation | 15 | 24.0 | 0.0351 ± 0.0011 | 4.72‰ | **7.44 ± 0.78** | | | | |

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| Type | Time (min) | Temp. (°C) | $m$ (nccSTP $^4$He) | $^{He}F_{Tube}$ | $n_0$ (nccSTP $^4$He) | $\omega$ (nccSTP $^4$He) | $^{He}F_{Apatite}$ | Corrected $\ln(D/a^2)$ | Uncorrected $\ln(D/a^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| Experimental | 240 | 400.1 | 101.1 ± 2.7 | | *7.40 ± 0.78* | 110.0 ± 3.2 | 56.2% | -13.281 ± 0.076 | -13.70 |
| | 15 | 34.9 | 0.1096 ± 0.0031 | | *16.3 ± 1.7* | | | | |
| Permeation | 15 | 23.8 | 0.0757 ± 0.0022 | 4.67‰ | **16.2 ± 1.7** | | | | |
| Experimental | 10 | 350.3 | 14.88 ± 0.41 | | *16.1 ± 1.7* | 5.8 ± 1.9 | 57.2% | -14.92 ± 0.92 | -15.42 |
| | 15 | 39.5 | 0.0567 ± 0.0017 | | *7.06 ± 0.74* | | | | |
| Permeation | 15 | 24.0 | 0.0330 ± 0.0011 | 4.71‰ | **7.00 ± 0.74** | | | | |
| Experimental | 60 | 400.4 | 14.21 ± 0.40 | | *6.97 ± 0.74* | 18.1 ± 1.4 | 60.1% | -13.31 ± 0.31 | -15.50 |
| | 15 | 40.4 | 0.0866 ± 0.0026 | | *10.9 ± 1.1* | | | | |
| Permeation | 15 | 24.0 | 0.0508 ± 0.0016 | 4.71‰ | **10.8 ± 1.1** | | | | |
| Experimental | 120 | 300.2 | 7.65 ± 0.22 | | *10.7 ± 1.1* | 0.7 ± 1.2 | 60.2% | – | -16.50 |
| | 15 | 39.4 | 0.0291 ± 0.0011 | | *3.79 ± 0.41* | | | | |
| Permeation | 15 | 23.7 | 0.01758 ± 0.00077 | 4.68‰ | **3.76 ± 0.41** | | | | |

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